Ticket 1 Domain Understanding

CS 4273 – Capstone | Fall 2026

Group F: James Cunningham, Colby Frison, Shone George Kutty Renjan, Patrick Ha, Alan Vo

Mentor/Client: Micheal Coffman

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Project information

Field	Value
Course	CS 4273 – Capstone | Fall 2026
Ticket	Ticket 1 – Domain Understanding
Project	Pilots Guide to Medical Certification
Group ID	Group F
Team members	James Cunningham, Colby Frison, Shone George Kutty Renjan, Patrick Ha, Alan Vo
Product Owner	Colby Frison
Mentor/Client	Micheal Coffman
Date	8/27/2026

Project Domain

Newer pilots often lack understanding as to what an FAA medical certification requires, where to look, how to get one, and how to move through the application process. Additionally, many questions arise around disclosure of medical history, as the typical assumption is disclosure of certain information will end an applicant's career before it even begins.

Sub-problems in the domain

Some information that must be addressed within the project is:

- Where to find reliable information
- What information must be disclosed on the medical application and what is the result of disclosure
- What happens when an Aviation Medical Examiner (AME) cannot issue a certificate
- What medical terminology means in layman's terms

Goals

Our goal is to continue the development of the web platform from previous semesters that new pilots can find and **want to use**. This can be done with clear navigation, clearly explained medical content, and tools that help pilots in getting correct and reliable answers. The platform should also push applicants to get healthy prior to application, seeing doctors for concerns and checkups before pursuing the medical certification.

Key Questions and Answers

As a group, we landed on five main questions, some of which focus on the content in the domain and others focus on the web-applications. Following the question is the consensus reached through the individual research of all group members.

Q1. What is an FAA medical certificate?

An FAA airman medical certificate assures that an applicant meets medical standards created by the FAA. There are also classes for certification, including:

- First class - Airline pilots
- Second class - Commercial pilots
- Third class - Student pilots / recreational pilots

This distinction is important as being aware of the proper class to apply for makes expectations apparent. It drives a clear distinction between what the applicant wants to do and what medical standards they need to reach.

Q2. What medical information must applicants disclose and what happens after disclosure?

Applicants must disclose their medical history, which includes current prescription medications, and health professional visits within the past three years. If requirements are met during examination with the Aviation Medical Examiner (AME), the certificate will be issued immediately. Otherwise, the application can be deferred for further investigation or denied.

Q3. How does a new pilot obtain a certificate?

Applications must first check their medical history to see if their medication is on the FAA-approved list. Then they will fill out the online application through the MedXPress portal to take a questionnaire to acquire a Confirmation Number. Get in touch with an AME (Aviation Medical Examiner) to schedule an appointment with their Confirmation Number. Lastly, they will take the physical within 60 days of submitting their MedXPress form. Where if they pass will then be granted their medical certificate.

Q4. Where should pilots get accurate information, and what does "FAA-acceptable" mean?

There are a number of approved sources for getting accurate and reliable information on the certification including the FAA Medical Certification pages (link), the AME Guide (link), and through the MedXPress page (link) which can only be accessed through approved devices.

There are other sources like online forums or unofficial blogs, but these should be used carefully as the information may be outdated, misleading, or outright incorrect.

Finally, FAA-acceptable means that content is accurate, encouraging, and the platform has good UX principles. Additionally tools (like AI) must not invent new rules, and since work may be integrated with the FAA Office of Communication wording and accuracy is very important.

Q5. What did prior OU teams build, and what should this semester continue?

Previously during fall 2026 and spring 2026, OU students worked on this FAA project and developed several protypes concepts like AI-assisted searches, interactive graphics, and JSON files. The groups also evaluated the accuracy of the result that AI generated and presented their concepts to FAA CAMI leadership.

We should continue to work on finalizing the prototypes developed by the previous groups, improving the AI assisted search, interactive graphics, and JSON-based tools. We should focus on making the system more accurate, easier to use, and easier for pilots to understand.

Mentor/Client role and Involvement

Micheal will take on a mix of both client and mentor by assisting us in topics related to the FAA domain while communicating requirements for the system. He will conduct periodic check-ins ensuring we are headed on the right path during development while also clarifying content and UX within the web application. We will be interacting with him primarily through Microsoft teams, which allows us to share files, send messages, and conduct meetings all through the same service.

References

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